Fewer Draws, More Fun: Searching for Unbalanced Positions in Chess

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Abstract. Draw outcomes dominate both correspondence chess tournaments and competitions among top computer chess engines, diminishing their competitive and entertainment value. This study addresses the issue of draw dominance in computer chess, with direct implications for correspondence play. We investigate the problem of identifying alternative starting positions that lead to unbalanced games, where outcomes are less predictable even for state-of-the-art chess engines. Our approach focuses on searching, selecting, and evaluating positions —those with a 75% winning probability—as a means to increase decisive results. By analyzing these positions through extensive engine-based simulations, we aim to not only reduce draw rates but also introduce a framework for ranking and measuring the strength of the strongest chess engines.

1 Introduction

Chess attracts a dedicated audience that appreciates its depth, strategy, and nuances. However, when it comes to playing correspondence games[1] or watching computer chess tournaments, the experience can feel less engaging due to a high frequency of draws and openings with predictable outcomes [6].

Our goal is to explore ways to make these events more dynamic and enjoyable by identifying special, unbalanced starting positions that reduce draw rates and introduce more varied gameplay. While the study of unbalanced chess positions may seem like a niche endeavor, it has significant implications for both competitive engine testing and human-computer interaction. By analyzing the huge available game datasets we aim to uncover rare and valuable positions that can lead to richer, more decisive games. The appeal here is for chess enthusiasts and those interested in artificial intelligence [19]. Watching computer chess games

[1] Correspondence games are played by humans through mail, email, or dedicated websites, allowing for deep analysis usually supported by computers.

© IFIP International Federation for Information Processing 2025
Published by Springer Nature Switzerland AG 2025
M. Sugimoto et al. (Eds.): ICEC 2025, LNCS 16042, pp. 3–14, 2025.
https://doi.org/10.1007/978-3-032-02555-5_1

(CCG) can be fascinating because they often feature moves and strategies that are beyond human comprehension, revealing the depth and complexity of chess.

In a game of Chess there are three possible results: White wins, Black wins, or draw. The latter outcome is the most frequent between masters when they play long time controls[2]. The draw can result from: an agreement between the players, a stalemate, insufficient material to checkmate, the 50-move rule, a threefold repetition of the position, or a dead position where no legal sequence of moves can lead to checkmate[3].

The problem of too many draws in chess, particularly in correspondence chess where computers are extensively used, is an issue that affects the dynamics and appeal of the game [8], even if some drawn games are actually quite entertaining [18].

In fact, modern chess engines are very strong and can often find the best moves in most positions. Thus, both sides can defend accurately and avoid mistakes, which usually leads to a draw. The reliance on engines by correspondence players reduced the scope for human creativity and original play. Players may follow engine recommendations rather than exploring novel ideas, leading to more predictable and less exciting games. A high draw rate can also make the game less exciting for spectators. This can affect the popularity and viewership of chess events.

The predictability of outcomes when both sides play near-perfectly can reduce the drama; we need to increase unpredictability to increase entertainment [9]. Increasing unpredictability in CCG is also useful when comparing chess engines or for tiebreaks in their tournaments.

Some have proposed rule changes to reduce the number of draws, such as modifying the scoring system (e.g., awarding 3 points for a win and 1 for a draw), introducing no-draw zones in the early stages of the game, or altering the sequencing of moves [4].

Some argue that the high draw rate is an inherent aspect of chess, reflecting the game's balance and depth. They believe that the pursuit of perfection is part of what makes chess a profound and enduring game [8].

In summary, the problem of draws in chess stems from the near-perfect play enabled by engines. Addressing this issue requires a balance between maintaining the integrity of the game and enhancing its appeal to players and spectators alike.

Our Research Questions (RQ) are as follows:

RQ1 *How can we identify starting positions that minimize the draw rate in computer chess tournaments*?

We call "*nuggets*" these starting positions, because we found that they are rare and precious. We will show how we mine game repositories searching for nuggets using an engine like Stockfish.

RQ2 *What impact does the use of nuggets have on the rating and comparative evaluation of chess engines in tournaments*?

[2] https://www.chessgames.com/chessstats.html.

[3] See here the complete official rules of chess.

We will show and comment on the results of some competitions where we used the nuggets as starting positions.

This paper has the following structure: in Sect. 2 we describe some background research on the topic of draws in chess; in Sect. 3 we outline our approach; in Sect. 4 we expose some results; in Sect. 5 we discuss our results; finally, in 6 we draw our conclusions.

WC32/final, World Championship 32 Final

TD Pheby, Ian M. (IA)

Category 11					1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	GM	SIM	Score	Wins	SB	RG	Place
1	510692	SIM	Edwards, Jon	2525		½	½	½	½	½	½	½	½	½	½	1	½	½	½	½	1	9½		9	2	68.25	0	1
2	180109	GM	Lecroq, Michel	2568	½		½	½	½	½	½	½	½	½	½	½	½	½	1	½	1	9½		9	2	68	0	2
3	141625	SIM	Osipov, Sergey Adolfovich	2499	½	½		½	½	½	½	½	½	½	½	½	½	½	1	½	1	9½		9	2	68	0	2
4	390086	GM	Neto, Horácio	2567	½	½	½		½	½	½	½	½	½	½	½	½	½	½	1	1	9½		9	2	68	0	2
5	81224	SIM	Zajontz, Rainer	2500	½	½	½	½		½	½	½	½	½	½	½	½	½	½	½	1	9½		8.5	1	64.5	0	5
6	490165	GM	Akdag, Murat	2574	½	½	½	½	½		½	½	½	½	½	½	½	½	½	½	1	9½		8.5	1	64.5	0	5
7	300204	GM	Jónsson, Daði Örn	2559	½	½	½	½	½	½		½	½	½	½	½	½	½	½	½	1	9½		8.5	1	64.5	0	5
8	19094	SIM	Mendl, Manuel	2537	½	½	½	½	½	½	½		½	½	½	½	½	½	½	½	1	9½		8.5	1	64.5	0	5
9	89356	IM	Ulbig, Stefan	2416	½	½	½	½	½	½	½	½		½	½	½	½	½	½	½	1	9½	8	8.5	1	64.5	0	5
10	750087	SIM	Blitsko, Boris Mikhailovich	2509	½	½	½	½	½	½	½	½	½		½	½	½	½	½	½	1	9½		8.5	1	64.5	0	5
11	141530	SIM	Nekhaev, Andrey Ivanovich	2455	½	½	½	½	½	½	½	½	½	½		½	½	½	½	½	1	9½		8.5	1	64.5	0	5
12	130766	SIM	Michálek, Miroslav	2480	0	½	½	½	½	½	½	½	½	½	½		½	½	½	½	1	9½		8	1	60.25	0	12
13	81204	GM	Moll, Reinhard	2554	½	½	½	½	½	½	½	½	½	½	½	½		½	½	½	½	9½		8	0	64	0	13
14	141537	GM	Kochemasov, Andrey Leonidovich	2554	½	½	½	½	½	½	½	½	½	½	½	½	½		½	½	½	9½		8	0	64	0	13
15	82299	SIM	Schwetlick, Thomas	2470	½	0	0	½	½	½	½	½	½	½	½	½	½	½		½	1	9½		7.5	1	56	0	15
16	390491	GM	Pessoa, Francisco Azevedo	2524	½	½	½	0	½	½	½	½	½	½	½	½	½	½	½		½	9½		7.5	0	59.75	0	16
17	84833	SIM	Bock, Steffen	2562	0	0	0	0	0	0	0	0	0	0	0	½	½	0	½		½	9½		1.5	0	11.75	0	17

Fig. 1. Final standings of 32 World Correspondence Chess Championship, 2023

2 Background and Related Work

The frequency of wins, losses, and draws in chess varies depending on the skill level of the players, the time control, and the specific context (e.g., casual play, over the board or correspondence tournaments, online platforms). However, some general trends can be observed.

Over the board (OTB) with long playing times, at the Grandmaster Level approximately 40% of games end in a draw [11,14]. In recent World Championships the percentile was even higher [1]. This is because top-level players are highly skilled at defending and often play very accurately, making it difficult for either side to secure a win. The remaining games are split between wins and losses, with a slight bias toward the player with the white pieces, who has the first-move advantage. Typically, White wins around 30–35% of the time.

At the Amateur Level draws are less common than at the Grandmaster Level. This is because mistakes are more frequent at the Amateur level than the Grandmaster level. Moreover, at this level wins and losses are more evenly distributed than at the Grandmaster Level, with White still having a slight advantage, winning around 35–40% of the time.

Elite correspondence tournaments are especially plagued by high rates of draws. They has seen draw rates as high as 100% in 2017 and 97% in 2019

[8]. Figure 1 shows the final standings of the recent 32nd World Correspondence Chess Championship, where most games ended draw.

These statistics highlight the prevalence of draws in elite human competitions. Since draws are correlated with the ability of players, draws in CCG are even more frequent than in human competitions [12]. For instance, in 2017 a match between AlphaZero and Stockfish resulted in 72 draws over 100 games [15]. The 2015 TCEC Season 8 Superfinal between Komodo and Stockfish experienced an 89% draw rate.

On digital platforms like Chess.com[4] or Lichess[5], the frequency of wins, losses, and draws can vary based on the time control. Faster time controls (e.g. blitz) tend to have fewer draws (around 10–20%) and more decisive results due to time pressure and mistakes. Slower time controls see more draws (around 20–40%) as players have more time to think and avoid mistakes.

Across all levels, the overall distribution is roughly as follows [5]: draws are 30–40%; wins for White are 35–40%; wins for Black are 25–30%.

These percentages are approximate and can vary depending on the dataset and context. For example, in high-stakes tournaments, players usually play more conservatively, leading to more draws, while in casual games, players take more risks, leading to more decisive results.

2.1 Chess Entertainment

The attention on the problem of draws in (computer) chess has been posed by Haworth and Cazenave a few years ago [8]. The authors also reported on a proposal by an anonymous player (nicknamed Aloril) who defined a method to eliminate draws in a chess variant called Mobility Chess. Anbarci and Mehmet [1] addressed a similar problem for chess tiebreakers. These tiebreakers reduce the quality and entertainment value of the games, thus authors presented an AI-based method for objective tiebreaking. The method first measures the quality of players' moves by comparing them to the optimal moves recommended by a chess engine, and in case of a tie declares as the winner the player with the higher quality score. This approach aims at preserving the quality of the game but increasing the fairness. Towards the same goal, Brams and Ismail [4] worked on a variation of chess that allows double moves at the opening, and proposed a method called Balanced Alternation that proved to generate fairer matches. Such fairness is counterbalanced by a higher probability of forcing a draw with optimal play, which is not necessarily entertaining.

Other authors studied the impact of game strategies on the entertainment dimension of chess. In [9] the authors proposed a way to measure the entertaining impact of a chess game. The metrics is built on the number of possible moves and the length of the game and was successfully used to study chess variants.

Similar approaches have been proposed to study entertainment in board games in general. Yannakakis and Hallam [20] proposed a method to quantitatively measure some psychological factors, namely challenge and curiosity, and

[4] Chess.com game repository: https://www.chess.com/games.

[5] Lichess game repository: https://database.lichess.org.

how these contribute to human entertainment. They exploited neural networks to model player satisfaction, in terms of these two dimensions, and to study how this is influenced by different games and strategies.

The work of Halim et al. [7] is also worth mentioning. The authors proposed a set of entertainment metrics tailored for platform games and, in order to test these metrics, implemented an evolutionary algorithm able to generate game rules using these metrics as objective functions. The generated games proved to be increasingly entertaining according to the feedback provided by human players.

Primanita et al. [13] studied uncertainty in games, and in particular how it affects not only the way a game is solved but also the way a game is perceived and enjoyed by the player. Experimental results demonstrated how uncertainty plays a key role for both these aspects and in particular for increasing the entertainment value of a game.

Concerning related works on uncertainty and win probability as opposed to traditional evaluation, there are not so many works. Recently, the "fragility" of traditional positional evaluation has been discussed in [3]. The idea of using win probability to improve position evaluation of chess games has been also studied in [17].

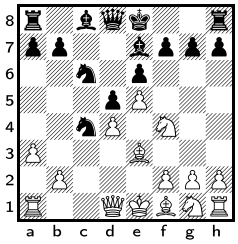
3 Method

The key idea of our proposal is to look for *unbalanced positions*, i.e. positions that are more likely to produce a White or Black win, thus reducing draw probability. We called these positions *nuggets*. In order to identify nuggets we exploit the probability models provided by Chess engines.

Stockfish (as well as many other engines) can provide a numeric evaluation of a position in units called *centipawns* (1/100 of a pawn). Newer versions of Stockfish include a built-in Win-Draw-Lose (WDL) probability model, which was introduced by Leela[6]. We also use ShashShess [10]. This program is derived from Stockfish and is enriched with the Shashin theory of chess as a complex system [16].

We have used WDL to study (unbalanced) positions in games from human World Championships and high level tournaments, e.g. Zurich 1953. For instance, this is a position from the 2024 World chess championship.

Gukesh-Ding Liren WCC match 2024, 13th game. **1 e4 e6 2 d4 d5 3 ♘c3 ♘f6 4 e5 ♘fd7 5 ♘ce2 c5 6 c3 ♘c6 7 a3 ♗e7 8 ♗e3 ♘b6 9 ♘f4 c×d4 10 c×d4 ♘c4** According to Stockfish: Position evaluation is +100 centipawns; White is winning ±. The win probability is (75, 25, 0).



[6] https://lczero.org/blog/2020/04/wdl-head/.

The primary criterion for selecting unbalanced positions is based on a winning probability threshold of 75%. This threshold was chosen for several reasons:

- In games among expert players, many positions tend toward equilibrium, leading to a significant proportion of drawn games. A 75% winning probability ensures that a selected position exhibits a clear imbalance while still allowing competitive play.
- Positions with an extreme win probability (e.g., above 90%) might be trivially winning for one side, reducing their practical significance. Conversely, a lower threshold (e.g., 60%) might still lead to frequent draws. The 75% threshold strikes a balance between competitive play and decisive outcomes.
- Many engines and databases classify positional advantages in terms of centipawn evaluations. Empirical studies suggest that a 75% win probability roughly corresponds to an evaluation of +1.00 to +1.50, a range often considered a substantial but not yet overwhelming advantage [17].

In order to systematically identify and select unbalanced starting positions, we followed a multi-step process involving database analysis, heuristic filtering, and expert validation:

1. **Database extraction**: we analyzed game databases with master-level play, namely World Chess Championships (1165 games) and Zurich 1953 tournament of candidates to the WC (210 games).
2. **Engine evaluation**: Each position was evaluated using Stockfish to determine the win probability for both sides.
   Positions were evaluated on:
   - Win probability between 70% and 80% (to ensure a slight but meaningful imbalance). Then, in order to validate the effectiveness of the selected positions, we conducted a series of simulations and tournaments played with long times. We employed Stockfish to analyze the selected positions and assess their long-term playability. Simulations were run under multiple time controls. Each position was tested in at least 1,000 engine-vs-engine games, ensuring statistical significance. We recorded:
     - Win/draw/loss distribution.
     - Centipawn evaluation trends over multiple plies.
     - Variability in different engine settings (e.g., different NNUE models configured in Stockfish).
   - Positional richness (avoiding forced tactical sequences leading to immediate resolution).
   - Historical frequency (preference for positions that have been tested in practical play).
3. **Manual refinement**: chess experts reviewed the positions we found to exclude trivial or artificially imbalanced setups, ensuring practical relevance.

The heuristic filters work as follows:

- *Material balance heuristic*: we exclude positions with excessive material imbalance, which often lead to predictable results. We check that material differences do not exceed a minor piece or an exchange. However, we accept positions where sacrificed material is compensated by positional factors. Example: we exclude queen vs. rook + minor piece but allow two rooks vs. queen with active play.
- *Tactical complexity heuristic*: we select positions rich in tactical opportunities to avoid forced sequences leading to quick resolution. When we analyze the branching factor—the number of reasonable moves per position, we prefer positions where multiple plausible continuations exist. Example: we prefer positions where both players have several attacking and counterattacking options.
- *Positional balance heuristic*: we favor positions where both sides retain strategic possibilities. We try to identify positions with unresolved tensions, such as weak pawns or contested open files. Moreover, we try to avoid sterile positions with few remaining pieces. Example: we look for opposite-side castling scenarios with mutual attacking chances.
- *Draw avoidance heuristic*: we exclude positions prone to perpetual check or threefold repetition, by checking positions where cyclic move repetition is likely. We also use MonteCarlo simulations to identify forced perpetual check sequences. Example: Discard positions where a best defensive strategy always results in repetition.
- *Dynamic balance heuristic*: we try to favor positions where play remains fluid, avoiding strategic stagnation. We evaluate piece mobility to ensure both sides have useful moves, and avoid closed, static positions with no clear plans. Example: we prefer double-edged middlegames over locked pawn structures.
- *Sensitivity to errors heuristic*: we prefer positions where small inaccuracies significantly impact the result. We assess how minor mistakes shift evaluation scores (e.g., from +0.5 to -1.0). Also, we avoid technical endgames with little room for blunders. Example: middlegame positions where an inaccuracy in defense triggers an immediate attack.
- *Database heuristic*: we use historical game databases to identify nugget positions that have emerged in practical play. We analyze games between humans or between computers to find positions with win probabilities near 75% or 25%, then we select positions that have resulted in varied outcomes or have been subject to deep analysis. Examples: positions from classic grandmaster tournaments (for instance, we analyzed all games from 1953 Zurich Candidate Tournament) or unbalanced computer games, like the wins by AlphaZero over Stockfish [15].

Table 1 provides a useful reference for evaluating chess positions in a human-like way by integrating probability, positional assessment, and historical playing styles associated with players such as Petrosian (defensive), Capablanca (strategic), and Tal (attacking). It is based on Shashin theory [16] and maps WDL (Win-Draw-Loss) ranges, probabilistic evaluations (*win probability*), positional characteristics, and Informator symbols to specific types of positions.

Table 1. Win probability ranges named following Shashin theory [16]. Central rows, from Low Petrosian to Low Tal, except the Capablanca one, that represents forced draw, are those useful for nuggets.

WDL Range (W, D, L)	Shashin Position's Type	Win Probability Range	Informator Symbols	Description
[0,3], [0,4], [96, 100]	High Petrosian	[0, 5]	-+	Winning: a decisive advantage, with the position clearly leading to victory
[4, 6], [5, 8], [92, 95]	Middle-High Petrosian	[6, 10]	-+ \ -/+	Decisive advantage: dominant position and likely winning
[7, 9], [9, 12], [88, 91]	Middle Petrosian	[11, 15]	-/+	Clear advantage: a substantial positional advantage, but a win is not yet inevitable.
[10, 12], [13, 16], [84, 87]	Middle-Low Petrosian	[16, 20]	-/+ \ =/+	Significant advantage: strong edge
[21, 35], [0, 16], [65, 79]	Low Petrosian	[21, 35]	=/+	Slight advantage with a positional edge, but no immediate threats
[0, 15], [40, 60], [40, 60]	Chaos: CapablancaâĂŞPetrosian	[25, 50]	↓	Opponent pressure and initiative: defensive position
[33, 33, 33]	Chaos: CapablancaâĂŞPetrosianâĂŞTal	[50, 50]	∞	Total chaos: unclear position, dynamically balanced, with no clear advantage for either side and no clear positional trends
[40, 60], [40, 60], [0, 15]	Chaos: CapablancaâĂŞTal	[50, 75]	↑	Initiative: playing dictation with active moves and forcing ideas
[0, 100, 0]	Capablanca	[50, 50]	=	Equal position. Both sides are evenly matched, with no evident advantage.
[65, 79], [0, 16], [21, 35]	Low Tal	[65, 79]	+/=	Slight advantage: a minor positional edge, but it's not significant
[84, 87], [13, 16], [10, 12]	Middle-Low Tal	[80, 84]	+/= \ +/-	Slightly better, tending toward a clear advantage. The advantage is growing, but the position is still not decisive.
[88, 91], [9, 12], [7, 9]	Middle Tal	[85, 89]	+/-	Clear advantage: a significant edge, but still with defensive chances.
[92, 95], [5, 8], [4, 6]	Middle-High Tal	[90, 94]	+/- \ +-	Dominant position, almost decisive, not quite winning yet, but trending toward victory.
[96, 100], [0, 4], [0, 3]	High Tal	[95, 100]	+-	Winning: a decisive advantage, with victory nearly assured.

This method ensures that the selected unbalanced positions are both theoretically sound and practically relevant, offering a structured approach for testing the strength of computer players.

Win probability, within the WDL model, is defined as $W + D/2$. We define *nugget positions* as those where the probabilities of winning and drawing are equal (e.g., $(W, D, L) = (50, 50, 0)$ for $winProbability = 75\%$, or $(0, 50, 50)$ for $winProbability = 25\%$). These positions exhibit maximum uncertainty, making the final result highly dependent on the quality of subsequent moves.

Positions with a win probability of approximately either 75% or 25% are particularly valuable because: they feature strategic or tactical imbalances that defy straightforward evaluation; they challenge engines by requiring deep calculations to assess the final outcome; they reduce the likelihood of games crystallizing into well-known theoretical draws.

4 Results

We have looked for nuggets positions in World Chess Championships WCC (1165 games) and in the Zurich 1953 tournament (210 games). We have found 12 nuggets in the former dataset and 15 in the latter. Nuggets in these datasets of games played by expert humans are very rare because humans tend to avoid uncertain positions, playing safely.

We then run two experiments to verify the impact of these nuggets on draws. In particular, we run two tournaments between different engines, in which all games were played over nuggets positions. We considered either engines with similar ELO strenghts and with different ones.

Table 2. A round-robin tournament (each match over 20 games) among different Stockfish variants of different strengths. Using 10 nuggets, each match was over 20 games, total 200 games. We got only 12 draws (6%).

	Program	Elo	1	2	3	4	5	**Total**
1	ShashchessDev	3500	-	18	20	20	20	**78/80**
2	AlexanderExpert	3190	2	-	18	19	18.5	**57.5/80**
3	AlexanderAdvanced	2299	0	2	-	17.5	19	**38.5/80**
4	AlexanderIntermediate	2199	0	1	2.5	-	18.5	**22/80**
5	AlexanderBeginner	1999	0	1.5	1	1.5	-	**4/80**

Table 2 shows the results of a tournament among five experimental engines - all derived from Stockfish - of very different Elo strengths. The tournament used 10 nuggets from WCC games. There were only 12 draws out of 200 games.

The second experiment, made by someone independent from the authors of this paper, is a tournament among eight strong engines having similar Elo ratings, with long thinking time (30 min with 10 s increment for every move), played over the 12 nuggets from WCC games. The pie diagram in Fig. 2 shows 40% of decisive results (white or black win). There are almost 60% of draw outcomes, because the engines are strongly optimized. However, over 672 games among strong engines (which usually draw more than 90% of their games), the decisive results are 268 (almost 40%). Full game scores from this tournament are available at CCRL Discussion Board - Golden Nuggets 16CPU.

These two experiments prove also that nuggets are useful to rank consistently the strengths of different engines. In the first case, when the players were of very different strengths, most outcomes were wins by the stronger players. In the second case, when the players were more evenly matched, the decisive results were numerous, and were consistent with the small differences in strength.

5 Discussion

The outcomes of the tournaments described in the preceding section show that nuggets effectively reduce the frequency of draws in computer chess tournaments

Players	8		
Nuggets	12		
duble round robin	168	games	
Total games	672		
White wins	267	39,73	%
draw	404	60,12	%
Black wins	1	0,15	%

Stockfish 20250213	115/168
ShashChess Dev	106.5/168
Dragon by Komodo 3.3	86.5
PlentyChess 4.0.9	85/168
Obsidian dev-15.04	80/168
Berserk 20250203	79.5/168
Caissa 1.21.7	71/168
Alexander 5.0	48.5/168

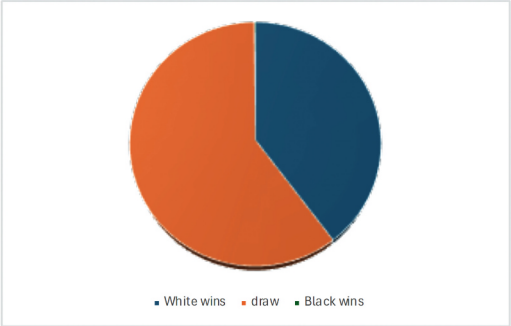


Fig. 2. Results of a tournament played among eight strong engines (12 nuggets from WCC, double round robin, 672 games). White wins + Black wins were 268 (almost 40% of total)

(RQ2). They introduce a controlled degree of uncertainty, making it less likely for games to follow well-established drawing patterns. The primary mechanisms behind this effect include (RQ1):

- Increased tactical and strategic complexity, making it harder for engines to reach quick, well-known drawn positions.
- A heightened likelihood of decisive results due to the natural instability of the selected positions.
- A greater dependence on precise calculation, leading to more engaging and dynamic games.

Incorporating unbalanced positions into computer or correspondence chess tournaments has several positive consequences. First, games become less predictable, so more engaging. By starting from positions where accurate evaluation and deep calculation are critical, double round robin tournaments allow to precisely assess reciprocal strength. Predefined openings often favor certain evaluation models: using these positions can mitigate the dominance of engines that excel in theoretical opening book knowledge.

Limitations

Despite the benefits, there are notable drawbacks in introducing tournaments based on nuggets. Certain engines may perform disproportionately well in specific types of unbalanced positions due to their evaluation functions. Identifying effective nugget positions requires extensive filtering to avoid those positions that resolve too quickly or lead to trivial outcomes. Last but not least, as engines improve, positions that were once considered chaotic or balanced may become easily winning or drawn.

6 Conclusion

We summarize our findings and the potential of unbalanced starting positions in addressing the high draw rates.

In standard chess competitions, final tournament standings and the resulting Elo rating are often influenced by factors such as win rates, opening preparation, and incremental advantages in symmetrical positions. However, these factors can sometimes obscure the true skill gap between players, particularly when a large percentage of games end in draws due to well-established theoretical lines or computational precision. By playing games from *nugget* positions we shift the focus from memorization and opening repertoire depth to practical decision-making, adaptability, and deep positional understanding. The expected result is a reduced impact of extensive opening preparation: since nugget positions are carefully selected to avoid theoretical main lines, players must rely on in-game evaluation rather than pre-existing knowledge.

In traditional settings, players of similar ratings may frequently draw against each other, making it difficult to distinguish who has superior skills. Nugget positions, by increasing complexity and decisiveness, ensure that even small skill differences manifest in results.

Nugget positions can be used for tiebreaks or experimental tournament formats. Moreover, nugget positions can serve as a new testing ground for engines aiming at training humans, measuring not just brute-force calculation but also strategic evaluation capabilities.

Using nugget positions in tournaments is a promising approach that pushes computer chess into less explored and more challenging territories. However, a careful selection and verification process is crucial to ensure that these positions remain balanced and engaging. A combination of WDL models, practical tests, and historical data analysis seems to be the best way to implement this idea.

Future research will explore:

- Search for new nuggets in openings repositories like the Encyclopedia of Openings [2].
- The effect of nugget positions on human play, analyzing whether they also reduce draws in human tournaments.
- A machine-learning approach to refine heuristic selection, using neural networks trained on engine evaluation shifts.
- An empirical study on whether these positions increase the rating gap between engines, better differentiating their strengths.

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